LESSON PLAN

BOMBS & COMPONENTS (32)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Basic requirements of an AV gun
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Basic requirements of an AV gun
5. Specific Objectives	32.1.1.1 Trainee will be able to explain requirement of air version guns.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AV gun components diagram; Safety: Do not handle live ammunition; Instructor supervision: Monitor trainee engagement and diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between a ground and air weapon?
(b) Why is weight a critical factor for aircraft armament?
(c) Can a standard ground LMG be fired from an aircraft without modification?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Air Version Gun • Weight reduction requirements • Recoil management systems • Mounting and feed mechanisms • Ammunition compatibility	L & Ds	PPT, IB	15 min
(b)	• Structural modifications for vibration • Cooling systems for sustained fire • Ejection port modifications • Traverse and elevation limits • Safety interlocks for flight	L & Ds	PPT, PDF	20 min
(c)	• Comparison of ground vs AV gun specs • Impact of G-forces on operation • Maintenance challenges in aviation • Future trends in AV armament • Summary of key requirements	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) List three specific modifications required to convert a ground LMG to an AV gun.
(b) Explain why recoil management is more critical in an aircraft than on the ground.
(c) Identify one safety interlock feature unique to air version guns.

12. Home Assignment and Follow Up	Prepare a brief note on the specific modifications carried out on a 7.62mm LMG to convert it into an AV gun, referencing the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Modification carried out on 7.62mm LMG to convert in to AV gun.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min